TURFY Sidecar Controller — Build Package v0.1

Sidecar retrofit for Rain Bird ESP-6Si irrigation controller. Raspberry Pi takes control of 6 zones + master valve through a transfer relay bank; any failure (power, crash, watchdog timeout) reverts to the Rain Bird with no software involvement. Sheets 1-4 attached.

Build rules (read first)

1. Both relay boards are standard 8-channel 5V opto-isolated boards with ACTIVE-LOW inputs and SPDT contacts (COM/NO/NC exposed per channel).
2. The 24VAC valve supply is tapped from the Rain Bird VT terminals. NEVER add a second 24VAC transformer to this design — the NC path back to the Rain Bird would parallel two out-of-phase secondaries.
3. Valve common field wire stays on the Rain Bird COM terminal. It is never moved or switched.
4. All logic grounds (Pi, relay boards, 555, opto emitters) meet at a single point. The 24VAC side floats — never bond it to logic ground.
5. Field wiring (24VAC, valves) on one side of the enclosure, logic on the other.
6. De-energized transfer bank must leave the Rain Bird wired exactly stock. Verify with continuity before first power-up.

Channel map

Ch	Zone bank relay	Transfer NC (from)	Transfer NO (from)	Transfer COM (to)
1	K1: COM=24VAC_HOT, NO->XFER1 NO	Rain Bird ST1	Zone K1 NO	Valve field wire 1
2	K2 (same pattern)	Rain Bird ST2	Zone K2 NO	Valve wire 2
3	K3	Rain Bird ST3	Zone K3 NO	Valve wire 3
4	K4	Rain Bird ST4	Zone K4 NO	Valve wire 4
5	K5	Rain Bird ST5	Zone K5 NO	Valve wire 5
6	K6	Rain Bird ST6	Zone K6 NO	Valve wire 6
7	K7	Rain Bird MV	Zone K7 NO	MV/pump field wire
8	spare	-	-	-

Raspberry Pi GPIO map (BCM)

GPIO	Signal	Direction	Notes
17	AUTHORITY	out	pull-down default; HIGH = request control
27	HEARTBEAT	out	idle HIGH; 50 ms LOW pulse every <= 10 s, from daemon main loop only
2 / 3	I2C1 SDA / SCL	bus	MCP23017 @ 0x20
5, 6, 13, 19, 26, 16	SENSE 1-6	in	H11AA1 outputs; LOW = Rain Bird station energized
20	SENSE MV	in	same
21	spare	in	reserved: flow meter pulses (edge IRQ)

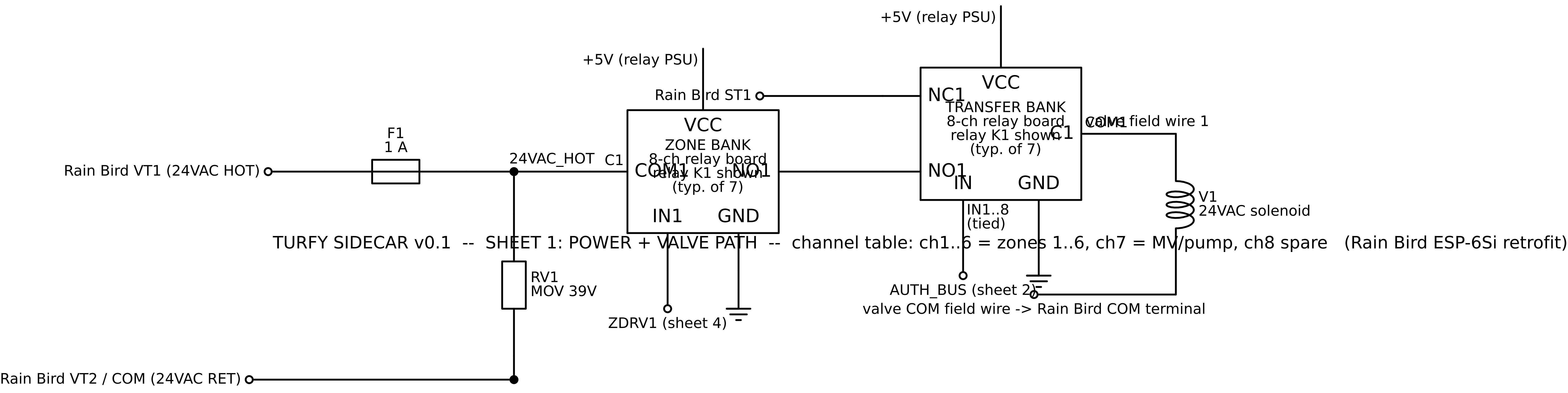
Bill of materials

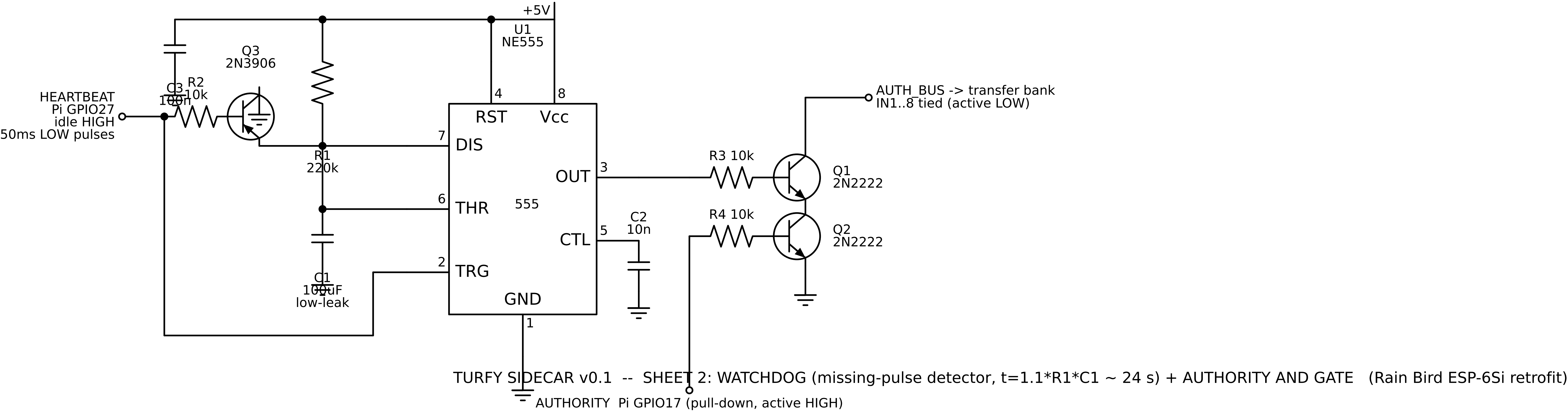
Ref	Qty	Part	Role
XFER, ZONE	2	8-ch 5V relay board, opto, SPDT, active-low IN	transfer bank / zone bank
U1	1	NE555 (DIP-8 + socket)	watchdog

Ref	Qty	Part	Role
U2	1	MCP23017 (DIP-28 + socket)	zone drive, I2C 0x20
U3-U9	7	H11AA1 (DIP-6 + socket)	AC sense opto
Q1, Q2	2	2N2222	authority AND
Q3	1	2N3906	watchdog cap discharge
R1	1	220k 1/4W	watchdog timing (t ~ 24 s)
R2, R3, R4	3	10k 1/4W	base resistors
Rs x7	7	2.2k 1W	H11AA1 LED current limit
Rp x7	7	10k 1/4W	sense pull-up to 3.3V
Rr	1	10k 1/4W	MCP23017 RESET pull-up
C1	1	100 uF low-leakage electrolytic	watchdog timing
C2	1	10 nF	555 CTL bypass
C3	1	100 nF	555 Vcc decoupling
Cf x7	7	1 uF	sense output smoothing (120 Hz)
F1	1	1 A fuse + holder	24VAC hot tap
RV1	1	MOV, 39 V clamp (e.g. V39ZA05)	transient clamp across VT
PSU	1	5 V / 2 A supply	relay coils (7 coils ~ 500 mA)
-	-	DIN enclosure, terminal blocks, hookup wire	assembly

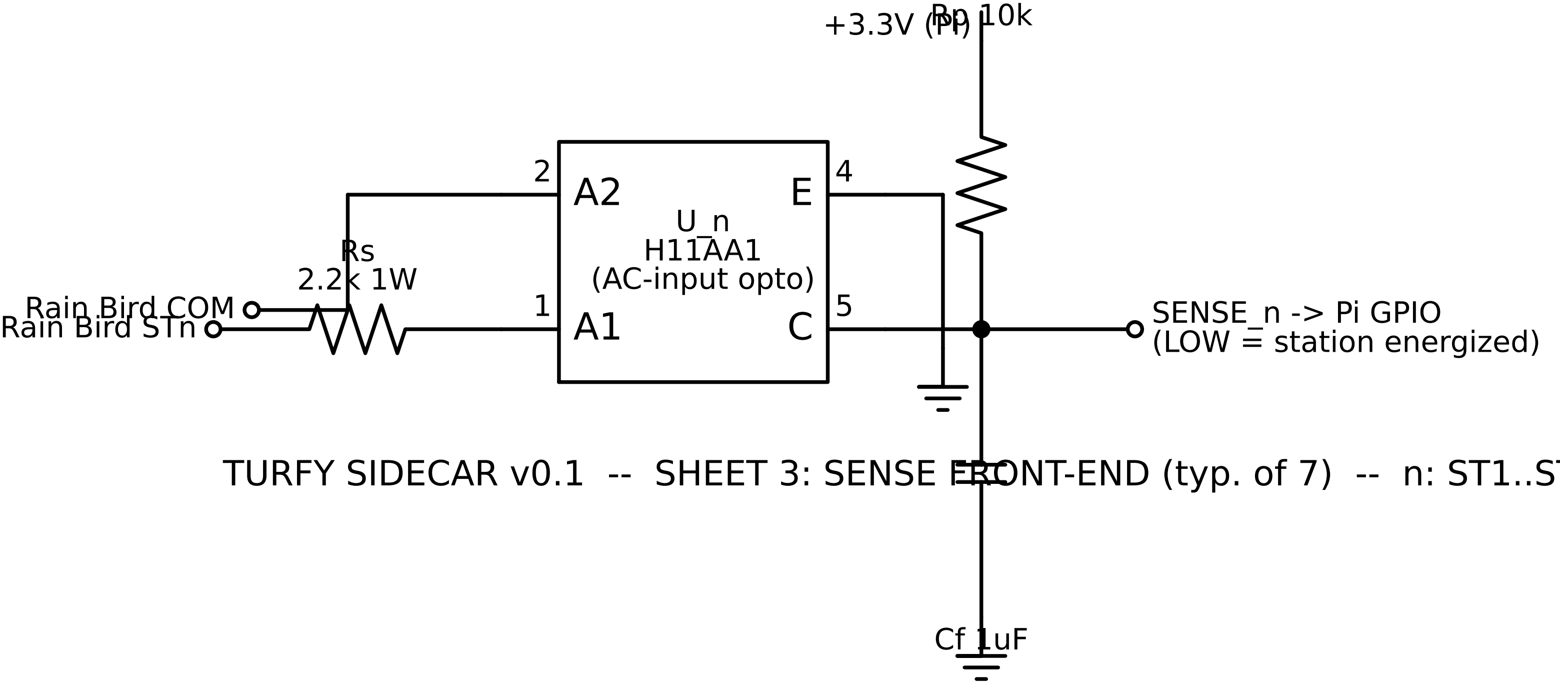
Acceptance test (before connecting valves)

1. Transfer bank unpowered: continuity ST_n -> valve terminal n for all 8 channels (Rain Bird path intact).
2. Power logic only (no 24VAC): boot Pi, verify all relays stay silent through two full reboots (GPIO chatter test).
3. Start daemon; confirm transfer bank engages only after AUTHORITY high AND heartbeat running. Kill -9 the daemon: transfer bank must drop within ~24 s.
4. Pull heartbeat wire with daemon running: same result.
5. Apply 24VAC, dummy load (24 V lamp or spare solenoid) on channel 1: verify Rain Bird manual start drives it with authority off, and MCP23017 bit 0 drives it with authority on.

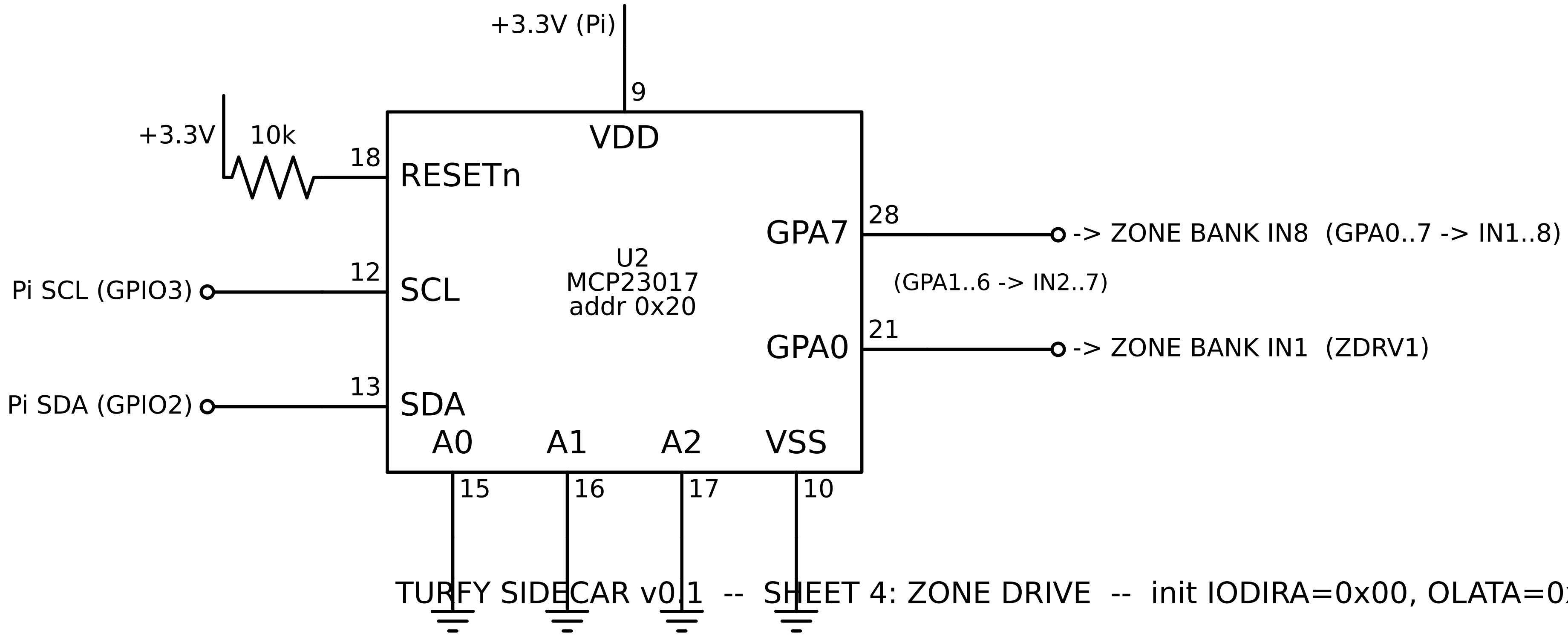

+5V (relay PSU)
+5V (relay PSU)
Rain Bird ST1
VCC
NC1
TRANSFER BANK
8-ch relay board
relay K1 shown
(typ. of 7)
C1
COM1
valve field wire 1
NO1
IN
GND
IN1..8
(tied)
V1
24VAC solenoid
F1
1 A
Rain Bird VT1 (24VAC HOT)
24VAC_HOT
C1
VCC
ZONE BANK
8-ch relay board
relay K1 shown
(typ. of 7)
COM1
NO1
IN1
GND
TURFY SIDECAR v0.1 -- SHEET 1: POWER + VALVE PATH -- channel table: ch1..6 = zones 1..6, ch7 = MV/pump, ch8 spare (Rain Bird ESP-6Si retrofit)
RV1
MOV 39V
AUTH_BUS (sheet 2)
valve COM field wire -> Rain Bird COM terminal
ZDRV1 (sheet 4)
Rain Bird VT2 / COM (24VAC RET)



TURFY SIDECAR v0.1 -- SHEET 2: WATCHDOG (missing-pulse detector, t=1.1*R1*C1 ~ 24 s) + AUTHORITY AND GATE (Rain Bird ESP-6Si retrofit)



TURFY SIDECAR v0.1 -- SHEET 3: SENSE FRONT-END (typ. of 7) -- n: ST1..ST6 -> GPIO 5,6,13,19,26,16; MV -> GPIO 20 (Rain Bird ESP-6Si retrofit)



TURFY SIDECAR v0.1 -- SHEET 4: ZONE DRIVE -- init IODIRA=0x00, OLATA=0xFF; one bit low at a time (software interlock)   (Rain Bird ESP-6Si retrofit)

active LOW = valve on; POR: all pins hi-Z -> relays OFF